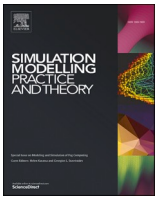
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Dynamic evacuation route planning for urban residents in flood disasters

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ABSTRACT

In recent years, flood disasters have frequently impacted urban areas, resulting in significant casualties and property loss worldwide. To ensure the safety of residents during flooding events, it is urgent to conduct large-scale evacuations in flood-prone areas. However, the impacts of flooding on urban evacuation are dynamically coupled with real-world conditions, which are often overlooked in current research. Therefore, a dynamic evacuation planning framework is proposed in this paper to provide safe and efficient evacuation routes for urban population under various flood scenarios. Firstly, the flood spreading process is accurately simulated across different rainfall patterns. Secondly, the dynamic influences of floods on travel speed, instability risk and road capacity are incorporated and quantified in the evacuation planning. Finally, the quickest evacuation routes are generated for evacuees in different residential areas to achieve both safety and efficiency objectives. Based on a case study, it is found that both the increased evacuation preparation time and larger rainfall intensity have an adverse effect on the evacuation safety and efficiency, significantly reducing the evacuation success rate under severe floods. The expressways carrying the largest traffic volume are crucial for ensuring the efficient evacuation during floods. Furthermore, the uneven utilization of shelters due to the flood impact and limited capacity should be considered in urban planning. The study can bring practical guidance for emergency departments and decision-makers to mitigate disaster losses and develop evacuation schemes in urban floods.

1. Introduction

Nowadays the trend of global warming has become increasingly severe, and the rapid development of urbanization has led to a sharp increase in impermeable underlying surfaces in urban areas [1,2]. As a result, extreme precipitation events now occur more frequently, leading to widespread flooding globally. Meanwhile, due to the limited capacity of urban drainage systems and the lack of efficient evacuation plans for residents [3], floods have become a high-frequency, large-scale, and highly destructive natural disaster

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all over the world [4].

According to the global natural disaster assessment report, floods are the most frequent natural disaster in 2024, with a total of 128 disasters occurring and affecting >48 million people. Based on China's statistics in the past 10 years, an average of 154 cities suffer from flooding annually, resulting in an average annual direct economic loss of 220 billion yuan [5]. The escalating urban rainstorm and flood disasters have irreversible and profound impacts on human survival and urban economy, posing a significant bottleneck restricting economic and social development. Therefore, it is crucial and urgent to carry out timely and effective evacuation for urban population to reduce disaster losses and ensure the safety of residents' lives. Besides, it is essentially important to develop urban flood evacuation models and optimize evacuation plans under various rainfall conditions to provide emergency decision support for large-scale evacuation in urban cities.

At present, extensive studies on evacuation under the context of flood disasters have been performed [6–12], however, the dynamic coupling impact of flooding on urban evacuation remains insufficiently addressed in existing research. In reality, the emergency evacuation during floods is a highly complex and information-intensive process. First, vehicles may face the instability risk in floodwaters, affecting the evacuation safety. Second, the travel speed of vehicles decreases under flooding conditions, reducing the evacuation efficiency. Third, severe floods lead to a decrease in road capacity, which can easily cause traffic congestion and ultimately reduce the traffic efficiency. More critically, these factors are interdependent and dynamically evolve with flood progression which needs to be comprehensively considered and accurately quantified in the urban evacuation process.

To address the aforementioned issues, this paper proposes a dynamic evacuation planning framework in flood disasters that considers flood hazard risks, road traffic conditions and vehicle movement characteristics. The dynamic optimal evacuation routes to safe shelters are finally generated for urban residents during flood emergencies. The remainder of this paper is organized as follows. Section 2 provides an overview of flood impact and route planning for urban flood evacuation in terms of methods and models. In Section 3, a dynamic evacuation planning framework is introduced in detail. First, the flood inundation process in urban areas is simulated by incorporating the pipe network data, river network data and terrain data, and flooding information such as flood depth and flow velocity is obtained. Second, the movement speed of vehicles in floods on urban roads is calculated, and the flood hazard for different types of vehicles is evaluated. Finally, optimal evacuation routes are generated based on safety and efficiency criteria. Section 4 presents a case study in Nanjing Luhe district. Section 5 compares the effectiveness of different planning methods, and analyzes evacuation times, evacuation failures, road traffic volumes and shelter utilization under various flooding scenarios. In Section 6, the main findings are summarized and the potential application for authorities and safety planners are discussed.

2. Literature review

2.1. Flood impact analysis

Vehicles are essential tools for long-distance travel and large-scale evacuation for residents, therefore the impact of flood on vehicle travel safety and efficiency on urban roads cannot be ignored, especially during emergency evacuation. Previous studies have demonstrated that the movement speed of vehicles will decrease with the increasing flood depth [13–16]. Moreover, the deterioration of traffic conditions caused by flooding leads to a significant decrease in road capacity and an apparent increase in road congestion, which further delays the evacuation efforts under flood disasters [17–19], especially in low-lying regions with poor drainage system. Besides, the road traffic safety risk in flooding events is also an important issue that poses a serious threat to the lives and property of residents. In severe circumstances, urban roads may become destroyed or submerged by floods, making it impassable for vehicles and seriously hindering the evacuation process. Therefore, to realize the safe evacuation, flood hazard on urban roads should be assessed in emergencies based on the following stages.

Firstly, the flooding information needs to be obtained from flood models. Currently, various hydrodynamic models have been developed by different institutes, which can be categorized as one-dimensional (1D) models, two-dimensional (2D) models, coupled one- and two-dimensional (1~2D) models, and three-dimensional (3D) models [20]. Secondly, to assess the hazard risk in the flood, researchers have established the instability functions for pedestrians and vehicles respectively based on theoretical and experimental methods [21–24]. For instance, Teo et al. [25] investigated the hydraulic behaviors of vehicles in urban floodplains through laboratory experiments in two hydraulic flumes. It was found that the threshold velocities for the initial movement of the car decreased with the increasing water depth. Xia et al. [26] obtained the criterion of vehicle stability in floodwaters for different orientation angles and ground slopes based on water depth and flow speed, providing a preliminary quantified criterion of hazard level for stationary vehicles in flood-prone urban areas. Arrighi et al. [27] used the relative submergence and the Froude number to assess the vulnerability of pedestrians and parked vehicles. However, the above studies only focus on the impact of hydrodynamic forces pertaining to static vehicles. To address vehicles in motion and endangered by floodwaters, Shah et al. [28] explored the governing effect of incipient velocity for moving vehicles by considering the rolling friction and driving force. It can be found that the water depth and flow velocity are the primary factors considered in the instability criteria in previous studies, providing a critical basis for urban flood risk assessment and evacuation decision-making in emergencies.

2.2. Evacuation route planning

The effective evacuation route planning is crucial to reduce the fatalities and mitigate the loss in flood prone areas [29]. At present, there are some studies on evacuation path planning under flood conditions. For instance, Fahad et al. [16] combined a traffic simulation model with a fluid dynamics model to construct a real-time evacuation planning framework, where the evacuation routes were

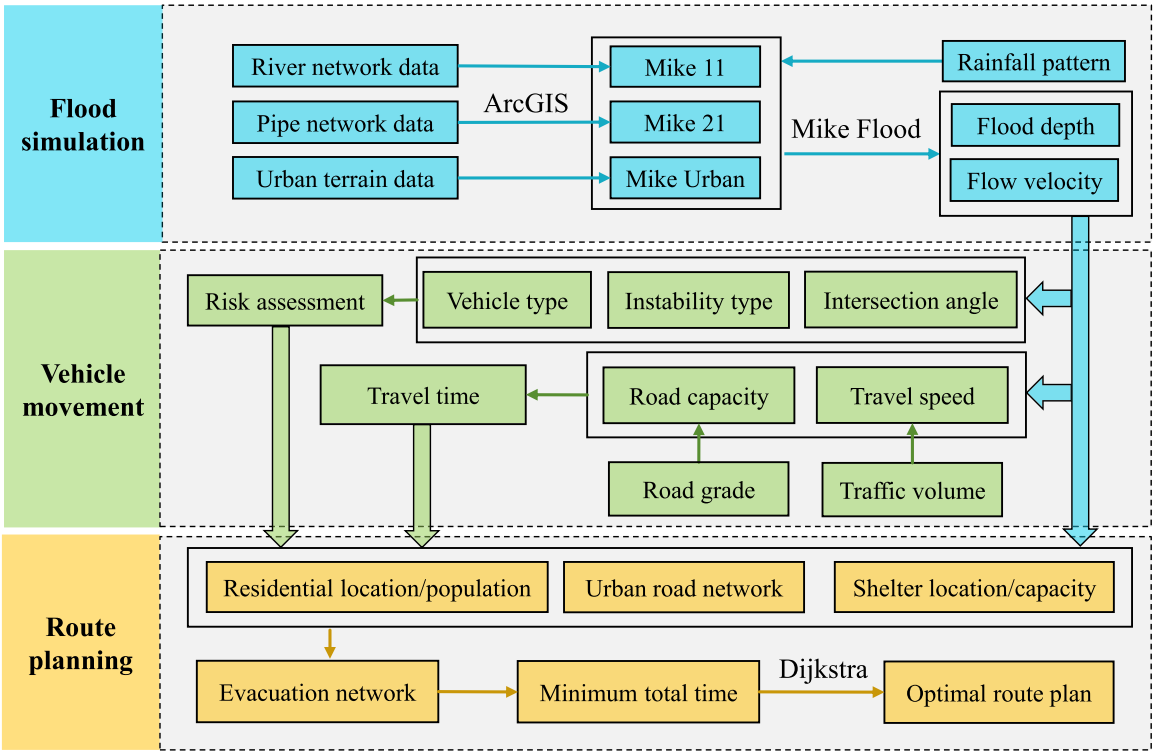


Fig. 1. Flowchart of the flood evacuation planning framework.

optimized based on the evacuation performance and traffic efficiency in flooding situations. He et al. [30] proposed a cellular automaton (CA)-based dynamic route optimization algorithm to simulate the flood evacuation in the suburb of Yangzhou city. The results demonstrated the superiority of the proposed algorithm over the traditional A* algorithm in terms of evacuation distance. Borowska et al. [31] investigated the effectiveness of self-evacuation by vehicle transport in flooded areas in Poland. Due to the dynamic evolution of urban flooding, it was found that the shortest paths in the pre-flood period were not necessarily be the optimal routes in the event of flooding. However, based on a multiagent evacuation model, Mizumura et al. [32] confirmed that the shortest route could be regard as a safe and optimal evacuation route under flooding conditions when the specific evacuation planning was not provided for inundated areas by local governments. Yazdani et al. [33] proposed an integrated decision model to facilitate the hospital transportation evacuation planning under emergency flooding circumstances. Based on the flood simulation results, the shortest paths were obtained for manager to minimize the total evacuation time. Besides, the proposed model can generate effective evacuation plans for the elderly by using multi-objective metaheuristic algorithms in extreme flood events [34]. Han et al. [35] used the Dijkstra algorithm to generate the shortest evacuation route for evacuees and the flooding risk was also evaluated based on the evacuation time predicted by the random forest algorithm. Finally, the vulnerable areas and times reducing the evacuation efficiency were identified in a case study. Yang et al. [36] developed an emergency evacuation route optimization model combining the ant colony algorithm by considering the search radius and road congestion, improving the selection efficiency of evacuation routes between transfer and resettlement villages. Although the above studies propose the emergency route plans based on different algorithms, the dynamic impacts of floods on vehicle motion, evacuation safety and shelter selection are not fully considered.

3. Methodology

When heavy rainfall leads to flooding in urban cities, it is essential to provide the real-time information on safe and efficient evacuation routes to residents in flood-prone areas. As shown in Fig. 1, the flood evacuation planning framework proposed in this paper is mainly divided into three components: 1) Simulation of flood inundation process in urban areas; 2) Analysis of vehicle movement in floods; 3) Optimization of evacuation routes based on traffic safety and efficiency.

3.1. Flood inundation simulation

In the field of hydrodynamics simulation, Mike models, known for their friendly interface and robust functionality, have been widely applied in urban rainstorm and flood simulations [37–39]. In this study, ArcGIS and Mike models are used collaboratively as follows: First, the river geometry and hydrological data for the study region are inputted into the Mike 11 to simulate the one-dimensional hydrodynamic processes of rivers and channels. Second, based on the urban road and building data preprocessed by

Table 1
Definitions and parameter values [44].

Parameter	Definition	Value		
		Unit	SUV	Car
u_c	Critical flow speed of sliding	m/s	—	—
u_f	Flow speed	m/s	—	—
h_f	Flood depth	m	—	—
α	Parameters related to vehicle shape,	—	0.367	0.492
β	tire type and road roughness	—	−0.451	−0.344
h_p	Vehicle height	m	1.737	1.48
l_p	Vehicle length	m	5.089	4.945
b_p	Vehicle width	m	1.983	1.845
G	Gravitational acceleration	m/s^2	9.8	9.8
ρ_p	Vehicle density	kg/m^3	203	170.44
ρ_f	Water density	kg/m^3	1000	1000
R_f	Ratio of critical depth to specific weight	—	0.551	0.65

ArcGIS, the Mike 21 surface flow model is employed to simulate the flood inundation in the two-dimensional urban surface. Simultaneously, the regional drainage network data are provided to the Mike Urban to simulate the flow within the pipe system and determine the inflow, flood points and overflow. Finally, the above simulation results from Mike 11, Mike 21 and Mike Urban are integrated into the Mike Flood platform to calculate the flood depth and flow velocity in urban areas.

3.2. Vehicle movement in floods

In the face of a sudden threat, self-evacuation is one of the most common self-preservation reactions [40]. In this study, the evacuation mode primarily involves the use of private vehicles including SUVs and cars by urban residents during flood disasters. The movement characteristics of these vehicles such as travel speed and stability state under flood conditions are analyzed in the following.

3.2.1. Travel speed

The movement of vehicles on urban roads is significantly influenced by the flooding conditions. Generally, the movement speed of vehicles will decrease with the increasing flood depth. Here the travel speed of vehicles affected by flood depth $V_{ij,f}^p$ is calculated as follows [41]:

$$V_{ij,f}^p = \frac{V_0^p}{2}\tanh\left(\frac{-h_{ij}+h_c^s/2}{b}\right)+\frac{V_0^p}{2} \tag{1}$$

where V_0^p represents the free travel speed of vehicle p under normal conditions without flooding, which equals to the designed speed of the road. h_{ij} is the average flood depth on the road section (i, j). h_c^s is the critical flood depth at which the vehicle comes to a complete stop, set to 40 cm for large SUVs a and 30 cm small cars respectively [42]. b indicates the decay elasticity coefficient, which quantifies how sensitive the vehicle speed is to changes in flood depth and is set to 4 in this study.

In addition to flood depth, traffic conditions can also influence vehicle speeds, further affecting the travel efficiency. Here the travel speed affected by traffic volume on the road $V_{ij,k}^p$ is considered as follows [13]:

$$V_{ij,k}^p = \frac{\alpha_1 \cdot V_0^p}{1+\left(q_{ij}/c_{ij}\right)^{\beta}} \tag{2}$$

$$\beta = \alpha_2 + \alpha_3 \cdot \left(q_{ij}/c_{ij}\right)^3 \tag{3}$$

where q_{ij} is the traffic volume on the road section (i, j). α_1, α_2 and α_3 are the regression parameters and their values are determined in the case study. c_{ij} represents the capacity of road segment (i, j). Based on the research findings from the "Highway Capacity Manual" (HCM) [43], the road capacity decreases by 2.01 %, 7.24 % and 14.13 % under the light rain, moderate rain and heavy rain conditions.

Finally, the travel speed of vehicle p on the road section (i, j) in the flood environment V_{ij}^p is determined as follows:

$$V_{ij}^p = \frac{V_{ij,k}^p}{2}\tanh\left(\frac{-h_{ij}+h_c^s/2}{b}\right)+\frac{V_{ij,k}^p}{2} \tag{4}$$

3.2.2. Flood hazard assessment for vehicles

Previous studies reveal that vehicle may lose stability in severe floods [26]. Based on the least favorable principle of flood vulnerability, the critical flow speed for vehicle sliding u_c is calculated as follows [26]:

Algorithm 1
Pseudocode of Dijkstra algorithm.

Input: Graph G with nodes V= (O, N, D) and edges E, the total number of vehicles M, current flood depth and traffic volume.
Output: The shortest time path from origin nodes O to destination nodes D.
Step1: Calculate the travel speed of vehicles on the road section in floods.
Step2: Calculate the travel time t of each path based on the travel speed.
Step3: Add all nodes to set Q (unvisited nodes)
for each vehicle p in M **do**
  While Q is not empty **do**
    Select node u in Q with min T
    Remove u from Q
    for each neighbor node v of u in Q **do**
Calculate current travel time on E (u, v): $t = t\,[u] + t\,(u, v)$
if $t < t\,[v]$ **then**
      Update the travel $t\,[v]$ to t
      Add the node v to path
end if
    end for
  end while
end for
Return path of all vehicles

$$u_c = \alpha\left(h_f/h_p\right)^{\beta}\sqrt{2gb_p\left[\rho_p h_p/\left(\rho_f h_f\right) - R_f\right]} \tag{5}$$

The definitions and values of the parameters are shown in Table 1.
Finally, the flood hazard index (FHI) is defined to quantify the danger level of roads under flood conditions as follows:

$$\mathrm{FHI} = u_f/u_c \tag{6}$$

When the FHI exceeds the threshold, vehicles face a high risk of instability in the flood and the corresponding roads are considered as dangerous and impassable in this paper (the threshold is set to 0.75) [44]. Besides, vehicles begin to float in floods when the flood depth exceeds the critical depth (0.67 m for SUVs and 0.45 m for cars) [26], similarly the roads are also regarded as dangerous for vehicles.

3.3. Optimal evacuation route planning

Residents need to reach their destination shelters safely and efficiently under emergency flood disasters. To prioritize safety, the dangerous roads assessed based on the flood hazard index are excluded from the evacuation route planning. In the following, the optimal evacuation routes minimizing the total evacuation time are obtained as follows.

$$\min T = \sum_{i\in I}\sum_{p\in\{P_i|i\in I\}} t_i^p\cdot\delta_i^p + \sum_{(i,j)\in E}\sum_{p\in\left\{P_{ij}|(i,j)\in E\right\}} t_{ij}^p\cdot\chi_{ij}^p \tag{7}$$

$$t_i^p = T_{i,b}^p - T_{i,a}^p \tag{8}$$

$$t_{ij}^p = l_{ij}\Big/V_{ij}^p \tag{9}$$

here the objective function T represents the minimum evacuation time consisted of travel times on intersections and road segments separately. $t_i^p$and t_{ij}^p indicates the travel time of vehicle p on intersection i and road section (i, j) respectively. V_{ij}^p represents the travel speed dynamically updated based on the real-time flood depth and the traffic volume on current section at each time step. $I = (O, N)$ is the set of origin nodes and intersection nodes. P_i and P_{ij} denote the set of vehicles on intersection i and road section (i, j). $\delta_i^p$is a binary variable that equals 1 if vehicle p is at the intersection i, and 0 otherwise. $\chi_{ij}^p = 1$ indicates that vehicle p is traveling on the road (i, j), and $\chi_{ij}^p = 0$ otherwise. $T_{i,a}^p$ and $T_{i,b}^p$ denotes the time when vehicle p enters and leaves the intersection i respectively. It is noted that if the number of vehicles exceeds the current intersection's capacity or the capacity of the upcoming road segment, the vehicle has to wait at the current intersection. l_{ij} denotes the road length between intersection i and j.

During the evacuation process, it is essential to ensure that the total number of vehicles remains constant as follows:

$$M = \sum_{i\in(O,N,D)} n_i + \sum_{(i,j)\in E} n_{ij} \tag{10}$$

where M is the total number of vehicles. n_i and n_{ij} represents the number of vehicles at the node i and on the edge (i, j) respectively.

Meanwhile, the traffic volume at each intersection must not exceed the capacity:

$$q_i \le c_i, i \in N \tag{11}$$

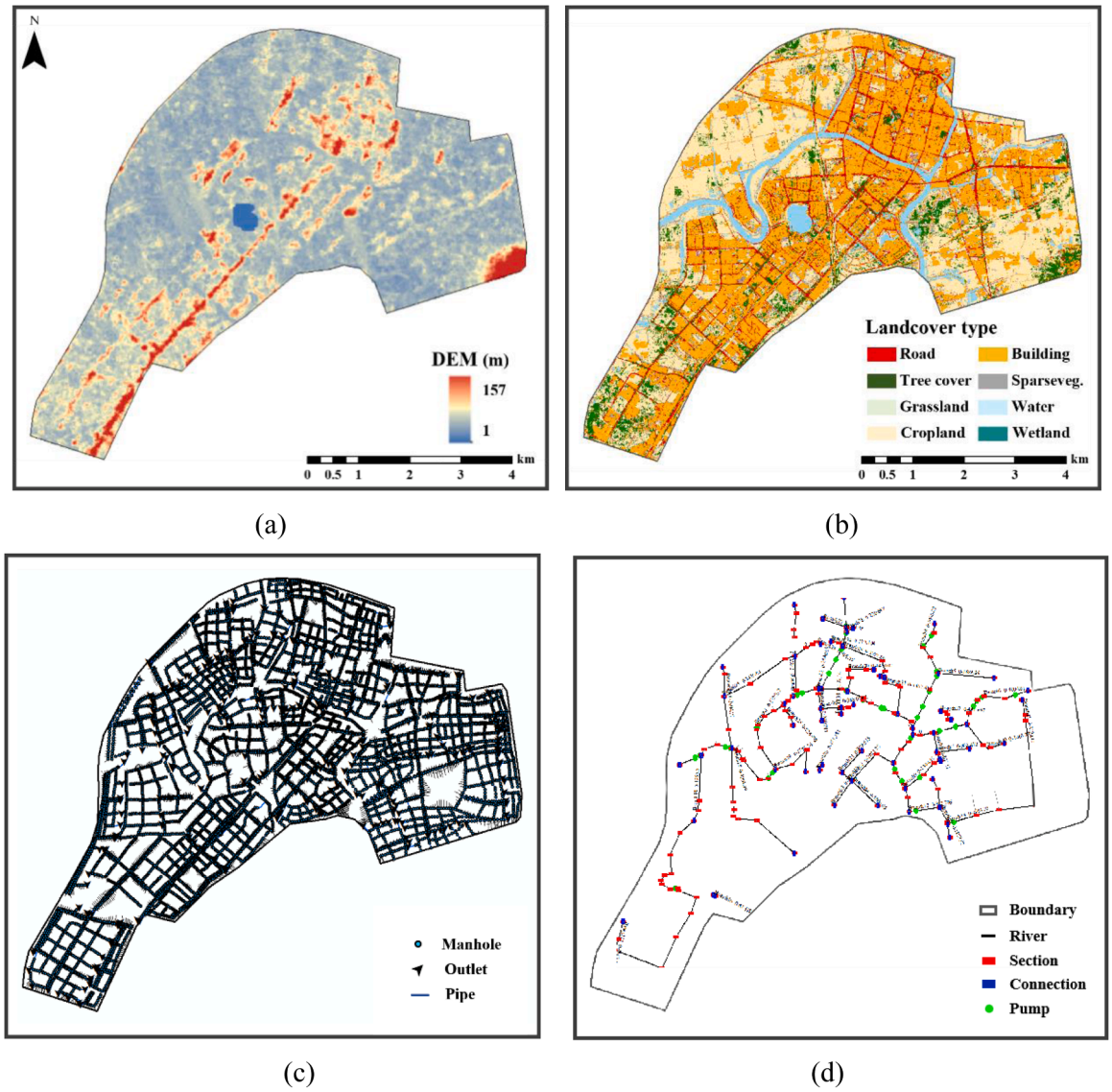


Fig. 2. Geographical information of the study area. (a) Surface elevation map, (b) landcover type map, (c) underground pipe network, (d) river network.

where q_i is the vehicle volume at the intersection i, and c_i denotes its capacity, which is set to 1800 veh/h/ln based on the HCM [43].

Similarly, the traffic volume on the road segment must not exceed the capacity:

$$q_{ij} \leq c_{ij}, (i,j) \in E \tag{12}$$

Besides, the number of residents accommodated in each shelter cannot exceed its capacity:

$$n_i \cdot q_c \leq c_i, i \in D \tag{13}$$

where q_c represents the passenger capacity of vehicle q. c_i denotes the capacity of shelter i.

Finally, the Dijkstra algorithm is employed to generate safe and efficient evacuation routes for all vehicles under flood conditions. It is noted that since flooding conditions are dynamic and can change in real-time, the optimal evacuation routes are replanned dynamically for vehicles in the intersection. The pseudocode of the route planning model is presented in Algorithm 1.

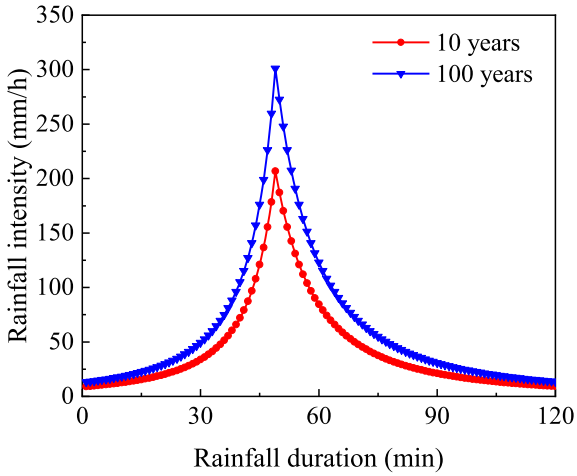


Fig. 3. Rainfall intensity in different rainfall patterns.

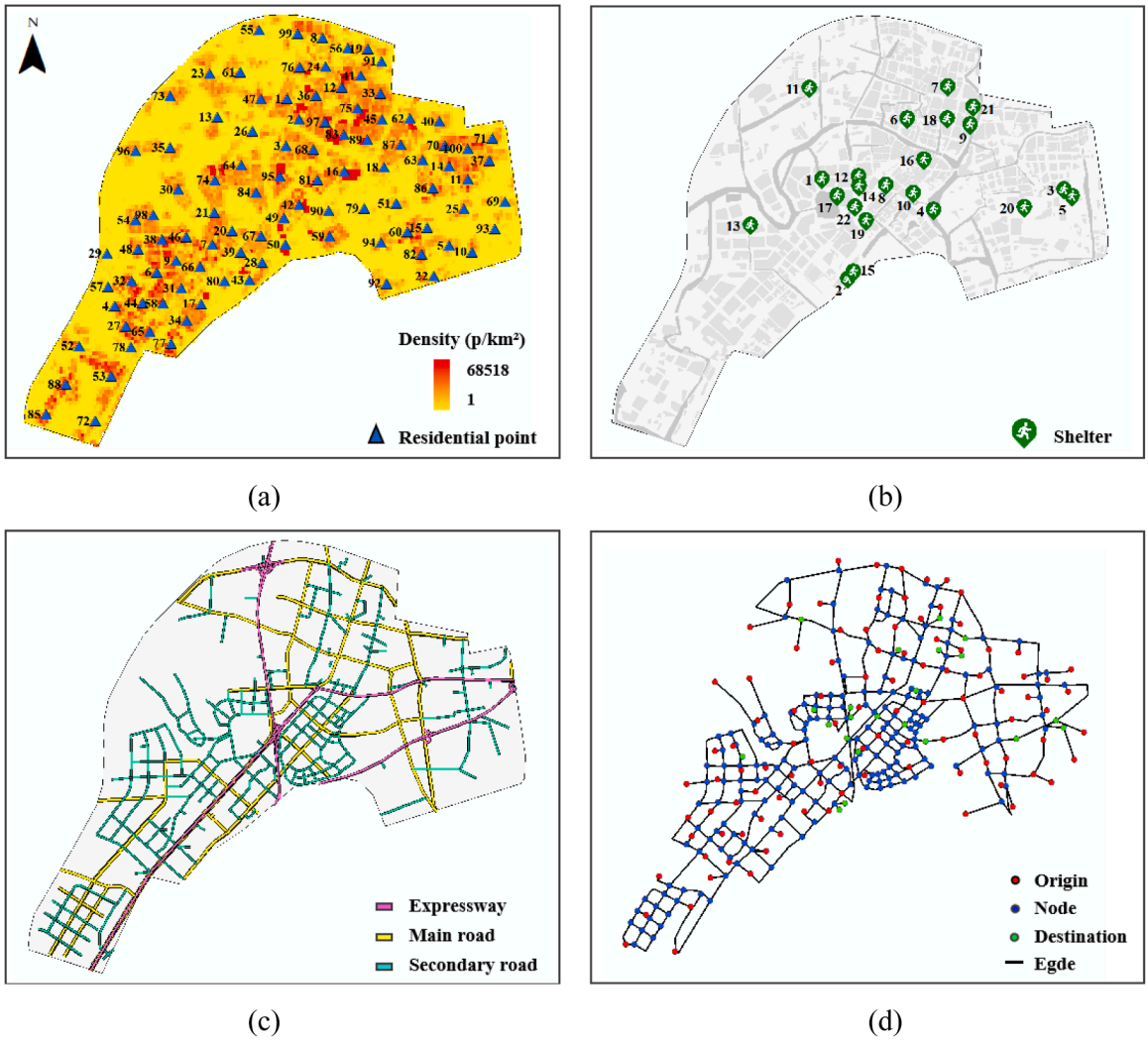


Fig. 4. Evacuation information of the study area. (a) Population density distribution and generated residential points, (b) Shelter location, (c) urban road network, (d) evacuation network.

Table 2
Parameter values for roads in the study area [13,48].

Value	Road grade		
	Expressway	Main road	Secondary road
Number of lanes	6	6	2
Designed speed (km/h)	80	60	40
Capacity (pcu/h/ln)	2000	1800	1650
α_1	1.00	1.10	1.40
α_2	1.88	1.89	1.88
α_3	4.90	4.85	6.97

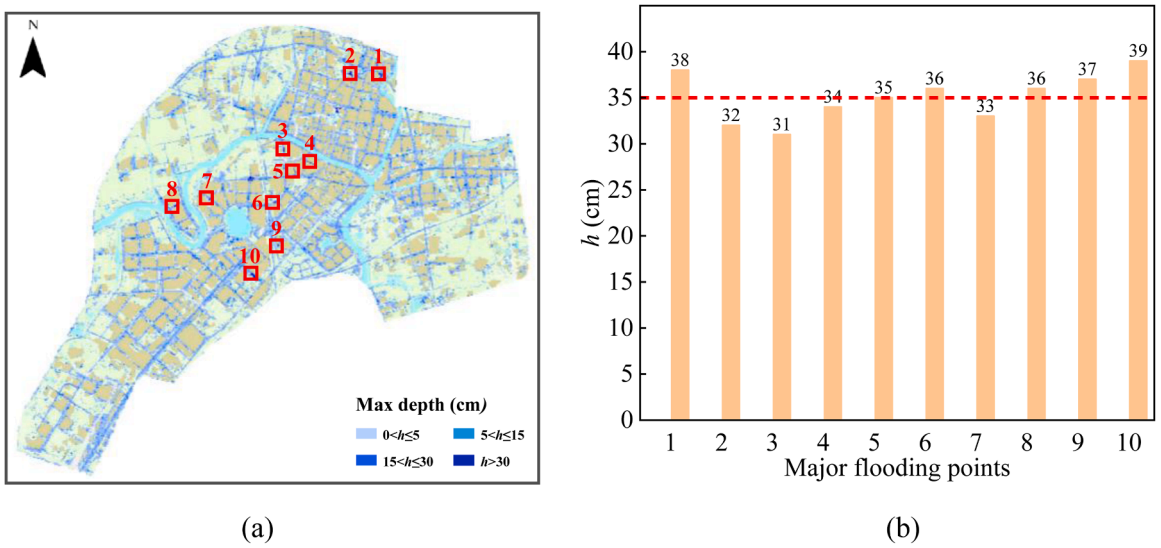


Fig. 5. Flood simulation validation. (a) Location of major flooding points, (b) Flood depth in the simulation and field measurement.

4. Case study

4.1. Study area and rainfall scenario

The selected study area is the subcentral area of Luhe District in Nanjing City, Jiangsu Province, China. The area experiences concentrated rainfall during the flood season from May to September each year and the average annual rainfall of about 1100 mm. As shown in Fig. 2, the urbanization rate in this region is high and the terrain is relatively low. In addition, the river network is densely distributed in the region, and the river channel capacity is insufficient, therefore the river is prone to overflow during rainstorm. Furthermore, the drainage capacity of the underground pipe system is reduced due to years of disrepair, and it is unable to timely and effectively discharge a large amount of rainwater, resulting in frequent urban waterlogging events in the area in history.

According to the "Nanjing Design Rainstorm Pattern Calculation Table" revised in April 2020, the Nanjing short-duration rainfall intensity is given as follows.

$$i_r = \frac{64.300 + 53.800 \lg R}{(t_r + 32.9600)^{1.011}} \tag{14}$$

where i_r is the rainfall intensity (mm/min), which can be used to distinguish the light rain (i_r<0.59mm/min), moderate rain (0.59<i_r<0.87mm/min) and heavy rain (i_r>0.87mm/min) conditions [45]. Here we mainly consider the heavy rain situations in this paper. t_r is the rainfall duration (min). R is the return period (a). In this paper, two rainfall patterns with a return period of 10 years and 100 years are designed as shown in Fig. 3. The total rainfall duration is set to 120 min and the rainfall peak coefficient is 0.4 based on the Chicago rainfall method [46].

4.2. Residential points

The total area is 73.5 km^2 and the total population is 162,224 in the study region. Based on the population density distribution in Fig. 4(a), the K-Means clustering algorithm is utilized to divide the study area into 100 residential points as shown in Fig. 4(a). The number of residents in each point is provided in Table A1 in Appendix. It is assumed that each vehicle is fully loaded, carrying 5 people

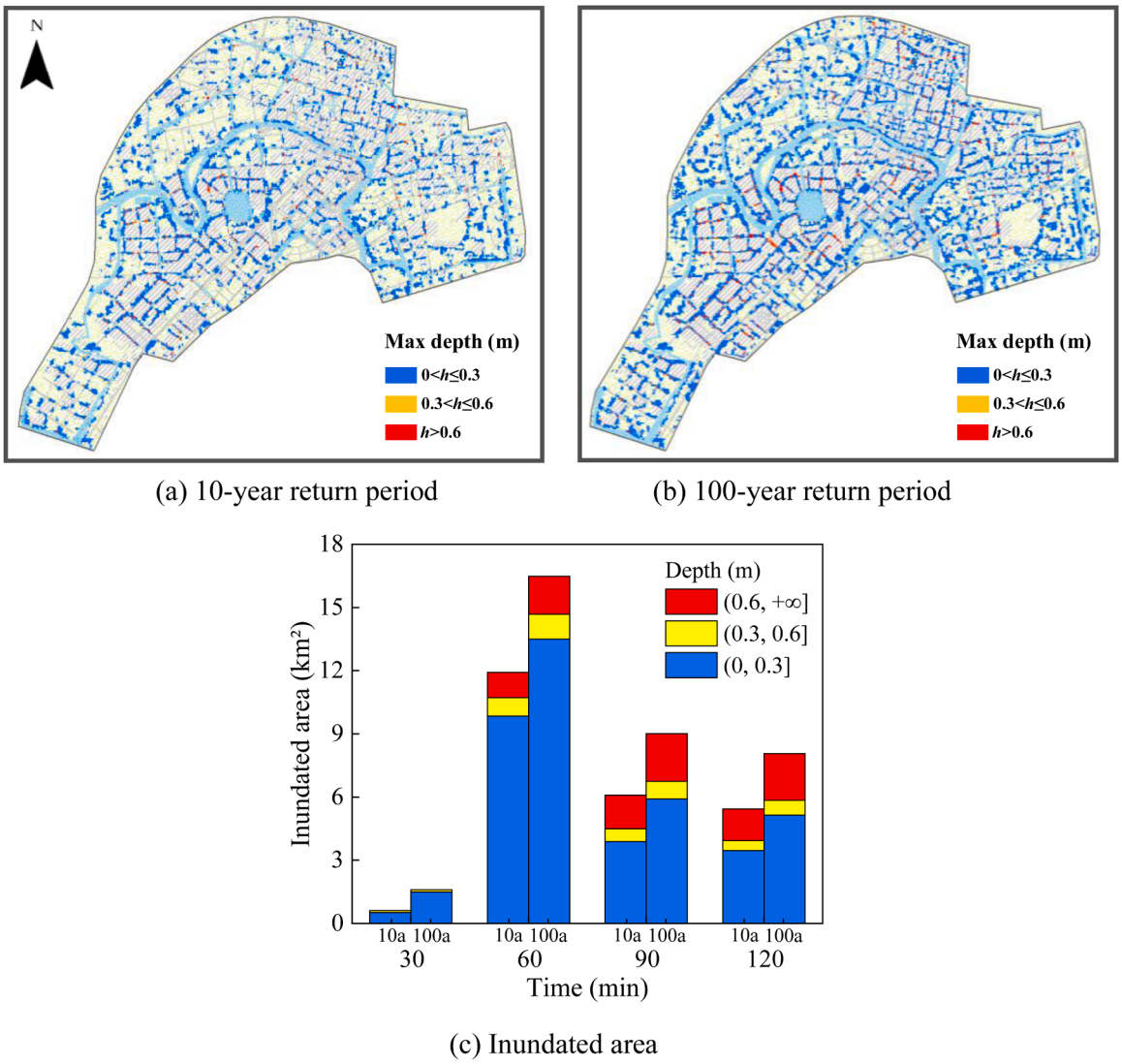


Fig. 6. Flood inundation simulation results in different return periods.

during the evacuation. Therefore, the number of vehicles to be evacuated are also determined in each residential point.

4.3. Available shelters

Emergency shelters are set up to provide the safety and basic needs of disaster victims, offering temporary accommodations in response to unexpected incidents. The detailed information of emergency shelters in Luhe District is published by the local government departments, which provides the geographical location, capacity and associated facilities of each shelter. Finally, 22 shelters are identified and displayed in Fig. 4(b) in the study area. Moreover, the capacity of each shelter is also given in Table A2 in Appendix.

4.4. Urban road network

Urban roads are the core link connecting various regions, and are essential for the effective evacuation route planning under emergency events. As depicted in Fig. 4(c), there are various types of urban roads in the study area. Here the expressways, main roads and secondary roads are mainly considered in the evacuation planning. Detailed information on roads is provided in Table 2.

4.5. Evacuation network construct

The residential points and shelters are designated as the evacuation origins and destinations, and the urban roads provide the

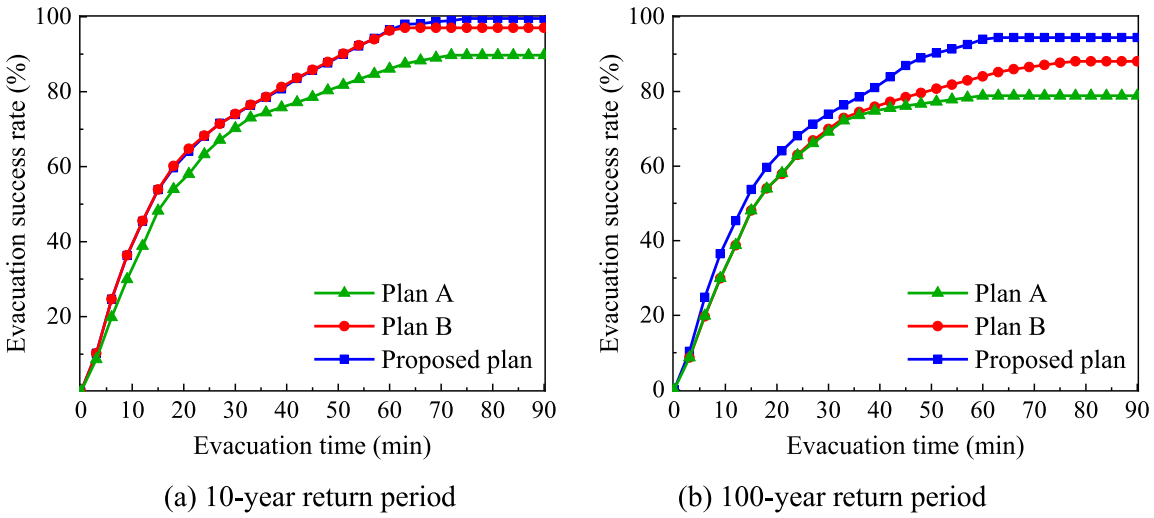


Fig. 7. Effectiveness comparison of different planning methods.

evacuation routes respectively. Finally, the evacuation network in Fig. 4(d) in the study area is constructed as follows:

$$G = (O, N, D, E) \tag{15}$$

Here G represents the evacuation network. O is the set of origin nodes, namely the collection of residential points. N represents the set of intersection nodes, which includes all road intersections. D is the set of destinations nodes, namely the collection of shelters. E refers to the set of edges connecting nodes, which contains all road segments in the study area. According to the vehicle ownership data from the Nanjing Municipal Bureau of Statistics, the ratio of driving SUVs to cars to evacuate is set at 3:7 in this study [47].

5. Result analysis

5.1. Validation of flood inundation simulation

The inundation result of floods will significantly affect the evacuation process and the final outcome. Therefore, to validate the reliability of the constructed rain-flood model, the rainfall data on June 19, 2023 were obtained from the meteorological station in Luhe District. According to the measurement results of the Water Affairs Bureau in Luhe District, the maximum water depth of 10 ponding points in Fig. 5(a) on that day was reported about 35 cm. Fig. 5(b) gives the corresponding simulation results in the above sites. It is found that the simulated water depths agree well with the measured data. The average error of water depth between the simulation and the measurement is 2.1 cm, proving that the constructed Mike models can accurately simulate the flood inundation process in the study area.

Fig. 6(a) and (b) display the spatial distributions of maximum water depth under 10 and 100 years return periods respectively. It can be seen that the severely inundated areas are mainly concentrated on the roads in the study area, with a water depth of >60 cm. The reason is that urban roads have high impermeability and low terrain, making them susceptible to be submerged in rainstorm events. Fig. 6(c) calculates the inundation areas with different water depths at different times. It is found that both the water depth and inundated area increase with the increase of return period. In addition, the rainfall intensity reaches its peak at 48 min and then begins to weaken as illustrated in Fig. 3. Therefore, with the increase of rainfall time, the flood inundation areas increase first and then decrease for the same return period.

5.2. Effectiveness of the proposed evacuation plan

Here different planning methods are compared to verify the effectiveness of the proposed method in this paper as follows: 1) The shortest route plan without considering flood risk (plan A), where the evacuation routes are determined solely based on the shortest road distance to shelters. 2) The shortest route plan with considering flood risk (plan B), where the dangerous roads are firstly excluded based on flood hazard risk, and then the evacuation routes are obtained based on the shortest distance from available roads to shelters. 3) The fastest route plan with considering flood risk (proposed plan in this paper), where the evacuation routes are generated based on the minimum evacuation time from safe roads to shelters.

The evacuation success rates of different planning methods under two rainfall patterns are shown in Fig. 7. It can be seen that the efficiency of plan A is relatively high in the early period of evacuation, especially under the 10-year return period scenario. The reason is that only a few roads have ponding, and the water depth and flow speed on urban roads are small as shown in Fig. 8(a) and Fig. 9(a). Therefore, the impact of flooding on traffic movement is tiny at this stage. However, in the later period, the ponding area on roads

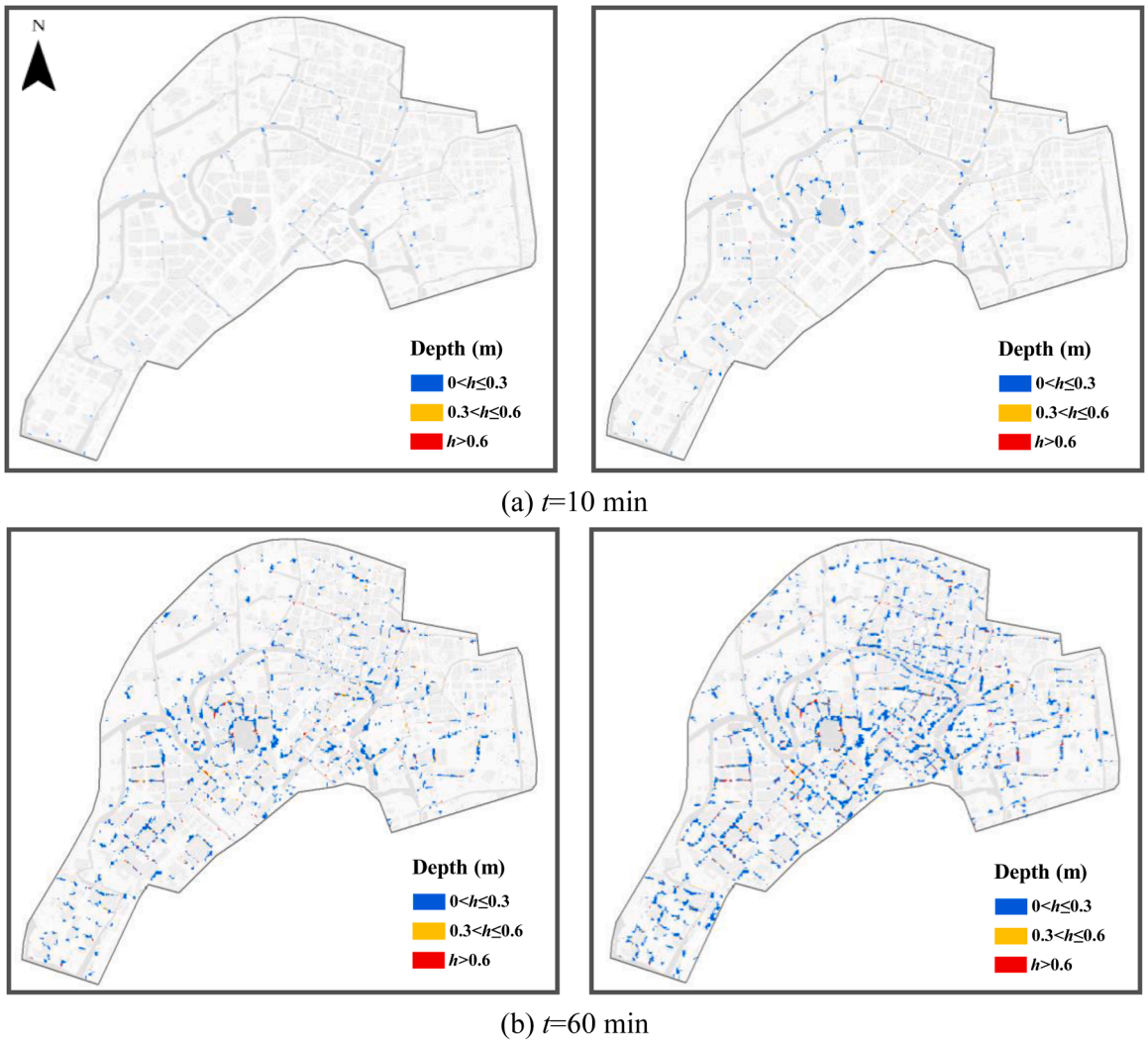


Fig. 8. Flood depth on urban roads. The left and right column represent the 10-year and 100-year return period respectively.

increases significantly and the flood depth and flow speed exceed 30 cm and 0.5 m/s at many areas in Fig. 8(b) and Fig. 9(b) respectively, especially in the 100-year return period, which seriously affects the traffic efficiency and safety. Due to the lack of consideration for flood hazard risk in plan A, vehicles are besieged by floods and unable to move, as a result, the evacuation success rate no longer increases under this circumstance. The final evacuation success rates are 89.8 % and 78.8 % in the 10-year and 100-year return periods respectively, suggesting that the ultimate evacuation performance of plan A is the worst, particularly in extreme flood events.

The evacuation process between plan B and the proposed plan is similar and the difference of evacuation success rate is also small in the 10-year return period. However, with the increase of rainfall intensity, the negative impact of floods on travel efficiency is becoming increasingly significant. Finally, the evacuation success rate of the proposed plan (94.4 %) is higher than that of plan B (88.1 %) in the 100-year return period. The above comparison highlights the superiority and effectiveness of the proposed planning method in ensuring a higher success rate under varying flooding scenarios.

5.3. Evacuation time and success rate

Previous studies have already demonstrated that most residents cannot evacuate immediately, requiring time to respond and prepare when facing disasters [49]. Fig. 10 displays the distribution of evacuation times for successfully evacuated vehicles under varying preparation times. Here lognormal, Gamma and Weibull function are used to fit the distributions, and the fitting coefficients indicate that the lognormal distribution provides the best fit. The positively skewed distribution implies that most residents are able to complete the evacuation activity in a short time period. The evacuation times for most successfully evacuated vehicles are <30 min, further demonstrating the efficiency of the proposed route planning method in this paper.

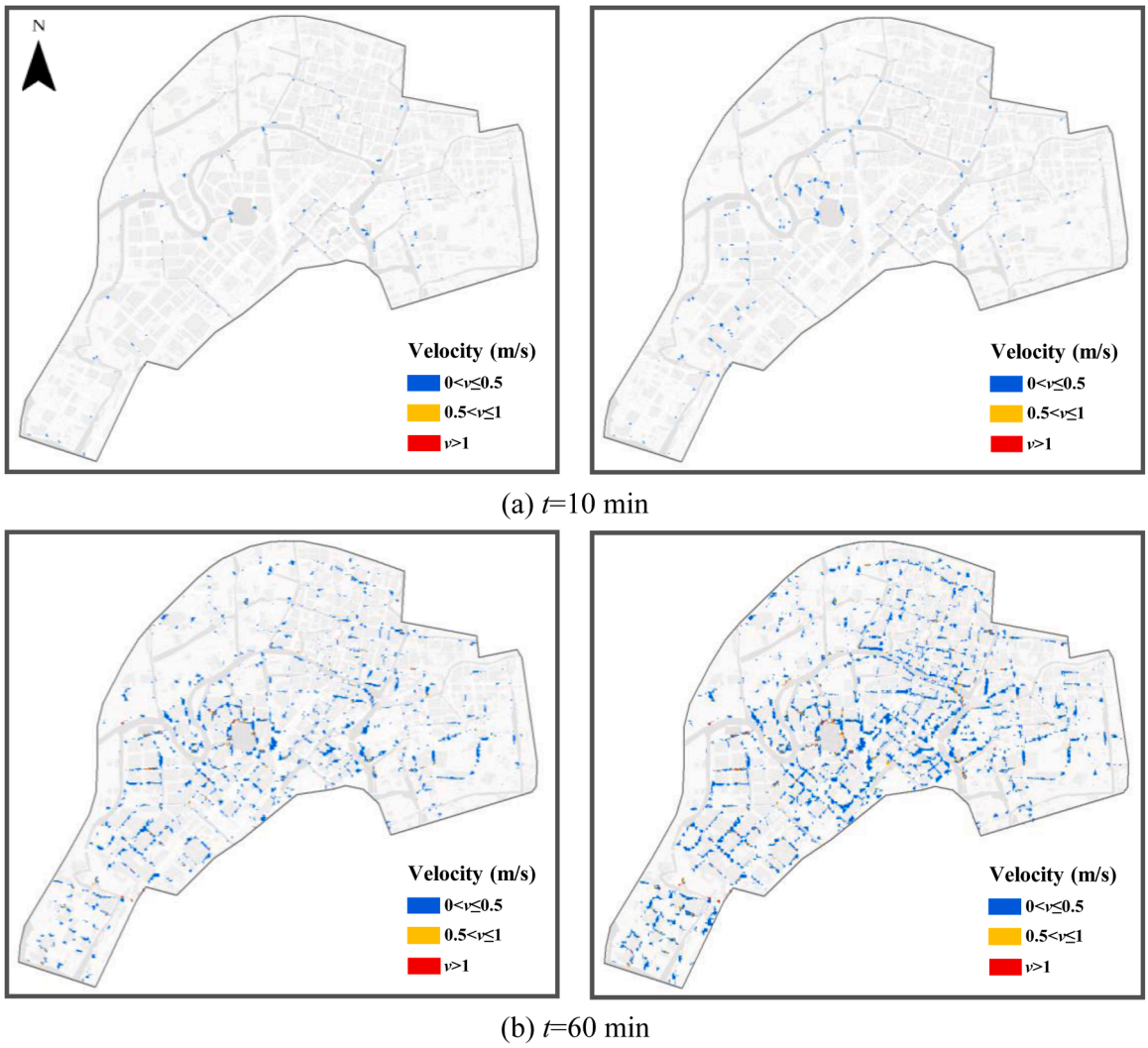


Fig. 9. Flow speed on urban roads. The left and right column represent the 10-year and 100-year return period respectively.

Fig. 11 provides the cumulative proportion of successfully evacuated vehicles under different preparation times. Specifically, the entire evacuation process can be divided into three stages: the early stage (0–30 min), the middle stage (30 min-60 min) and the later stage (>60 min). It is found that 80 % of vehicles can successfully arrive at the shelter in the early stage (except for the scenario of 30 min preparation time in the 100-year return period). As depicted in Fig. 3, the rainfall intensity is still low and the impact of flooding on road traffic is relatively small, therefore vehicles can move quickly during the early period. However, in the middle stage, the rainfall intensity increases and the influence caused by floods also increases, as a result, only a small number of vehicles can successfully evacuate at this time period. At the later stage, the evacuation situation has become extremely severe, and the majority of vehicles that have not yet evacuated are trapped by floods on the road and unable to reach shelters under this circumstance.

In Fig. 10, it is noted that the mean evacuation times of evacuated vehicles are smaller in the 100-year return period than that in the 10-year return period, which can be attributed to the significant variation in the final evacuation success rates across different cases in Fig. 11. It is found that the evacuation success rate decreases with the increase of return period. As displayed in Fig. 11(a), nearly all vehicles (99.5 %) can evacuate successfully within 90 min in the 10-year return period, however, with the increase of rainfall intensity, 5.6 % of vehicles fail to evacuate and cannot arrive at the shelters in the 100-year return period. As depicted in Fig. 12, the increase of preparation time further aggravates the situation. The evacuation success rate decreases with the increment of preparation time under different return periods. Under the 30 min preparation time scenario, the success rate decreases by 18.8 % and 24.6 % when compared to the prompt evacuation in the 10-year and 100-year return periods respectively. Due to the flood spreading during the preparation time period, the road capacity declines and the travel speed of vehicles also significantly decreases under severe flood disasters. As a result, the deteriorating traffic conditions make it increasingly difficult for vehicles to evacuate safely and quickly.

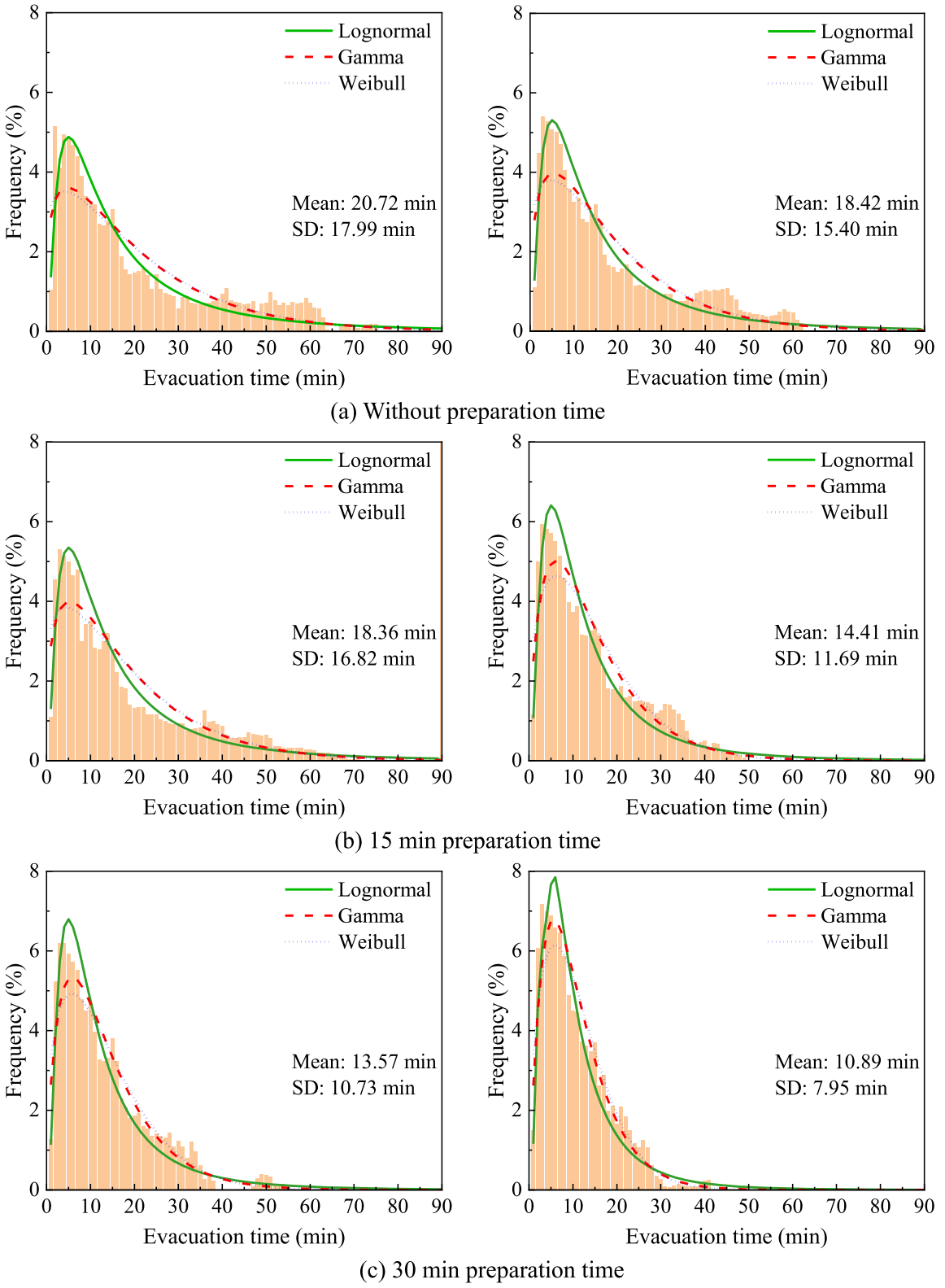
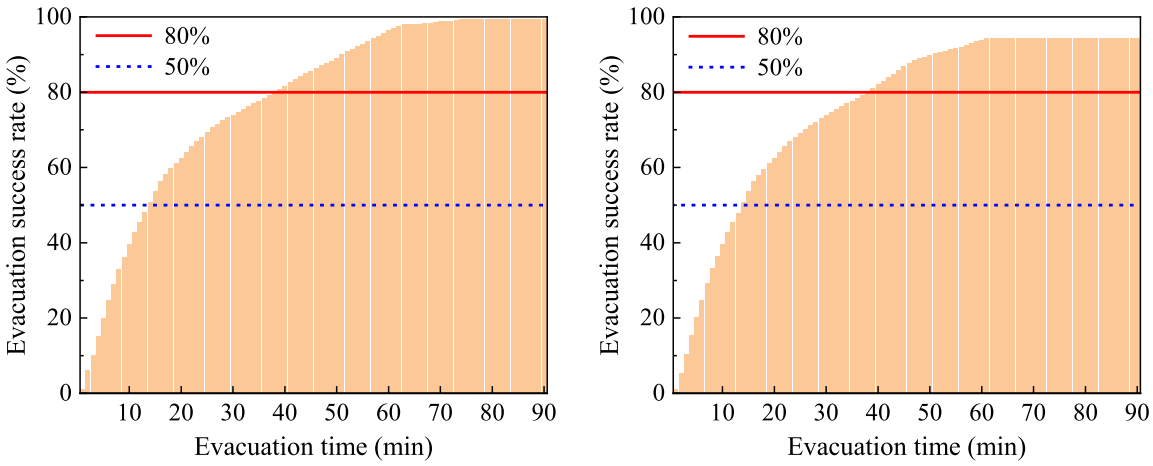
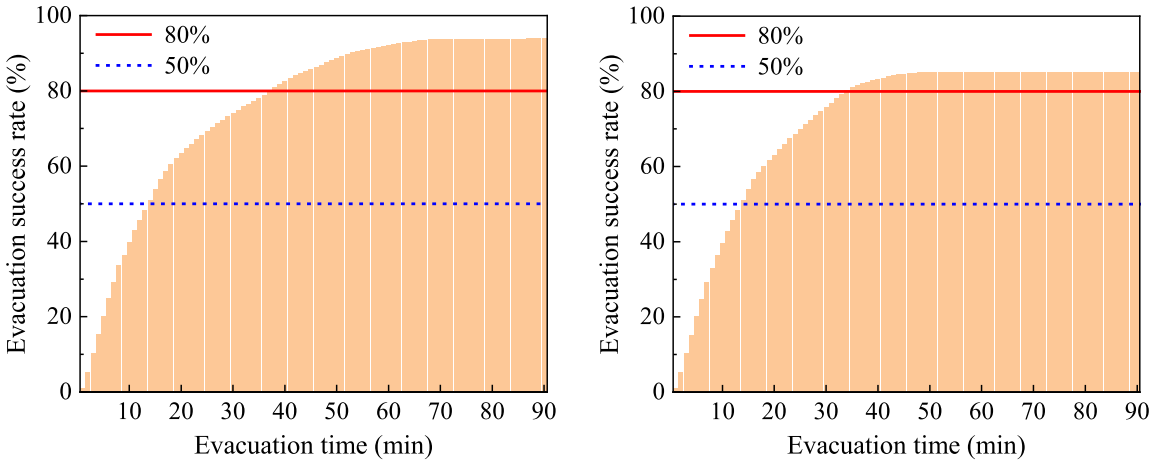


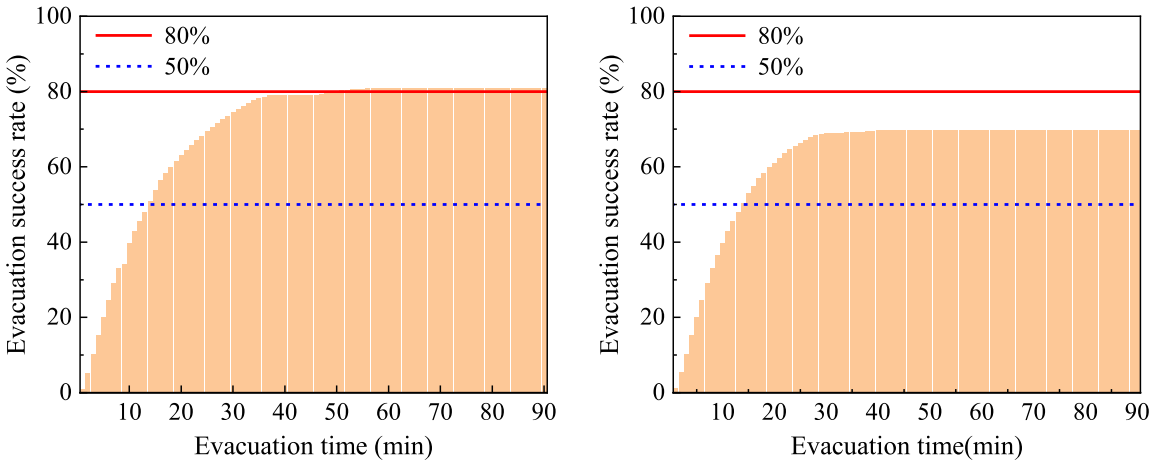
Fig. 10. Evacuation time distribution for successfully evacuated vehicles under different preparation times. The left and right column represent the 10-year and 100-year return period respectively.



(a) Without preparation time



(b) 15 min preparation time



(c) 30 min preparation time

Fig. 11. Cumulative proportion of evacuated vehicles under different preparation times. The left and right column represent the 10-year and 100-year return period respectively.

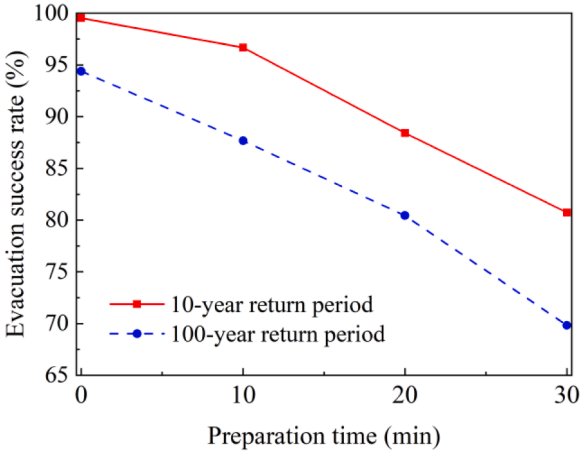


Fig. 12. Evacuation success rate under different preparation times.

5.4. Evacuation failure

The evacuation failure is investigated in this section to provide deeper insight into the whole evacuation process. Fig. 13 displays the final location of vehicles that were not successfully evacuated. It is found that flooding on urban roads is the primary reason of evacuation failure as it renders roads impassable. Based on the flood hazard value, the dangerous or impassable roads can be identified in real time. As shown in Fig. 13, there are only few dangerous roads under the 10-year return period without preparation time, while the number of dangerous roads increases significantly under the 100-year return period. Besides, the length of dangerous roads in the study region increases with the increment of preparation time under both return periods. This trend is understandable as more dangerous roads lead to more vehicle evacuation failures. Therefore, with the increase of rainfall intensity and preparation time, it is observed that the locations of evacuation failure vehicles extend from the expressway with the largest capacity to the secondary roads connecting to shelters. Furthermore, the evacuation failed vehicles also impede the moving of other vehicles on roads, resulting in severe evacuation delays and an increase in evacuation failures.

To figure out where the vehicles that failed to evacuate came from, the origin of evacuation failed vehicles is analyzed in Fig. 14. It is found that there are only six dispersed origins of failure in the 10-year return period without preparation time scenario, which are located at the margin of the study area. However, as both the return period and preparation time increase, the number of evacuation failure origins increases dramatically. Additionally, their locations shift closer to the central area and nearer to the periphery of shelters. More secondary roads are submerged by floods and some important roads connecting to shelters also become impassable in the 100-year return period with 30 min preparation time scenario. The capacity and efficiency of road network are seriously degraded in extreme flood events, which cannot provide safe and effective evacuation routes for residents.

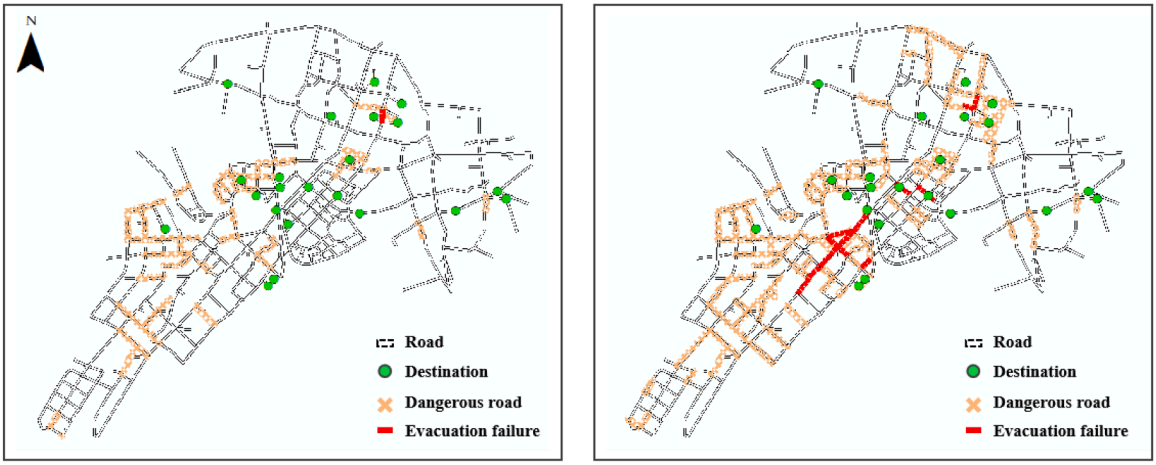
5.5. Road traffic volume

The traffic conditions on urban roads can influence the evacuation process. In Fig. 15, it is found that the distribution of traffic volume varies greatly on different types of roads. The expressway with the highest capacity carries the maximum traffic volume of nearly 5000 veh/h during the evacuation, which is crucially important for ensuring the overall evacuation efficiency. Even though the traffic volume on secondary roads is the smallest, those roads connecting with residential areas and shelters are also essential to ensure the final successful evacuation. However, no significant difference is found in traffic volume between different return periods and different preparation times.

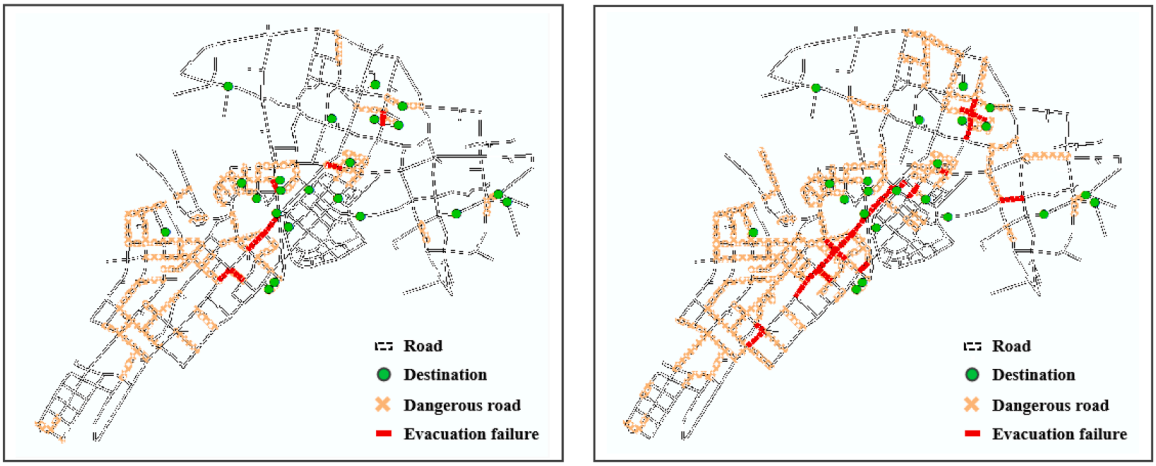
5.6. Shelter utilization

The ultimate usage rate of each shelter is analyzed in Fig. 16. It is shown that the shelter utilization is extremely unbalanced during the flood evacuation. Many shelters such as No 3, 5, 6, 11, 12, 16, 19, 20, 21, 22 are fully utilized while the utilization rate of other shelters such as No 17 and 18 are $<$20 % in both 10-year and 100-year return period. Moreover, the utilization rate of shelter No 9 is higher than 30 % in the 10-year return period while it drops to 0 % under the 100-year return period.

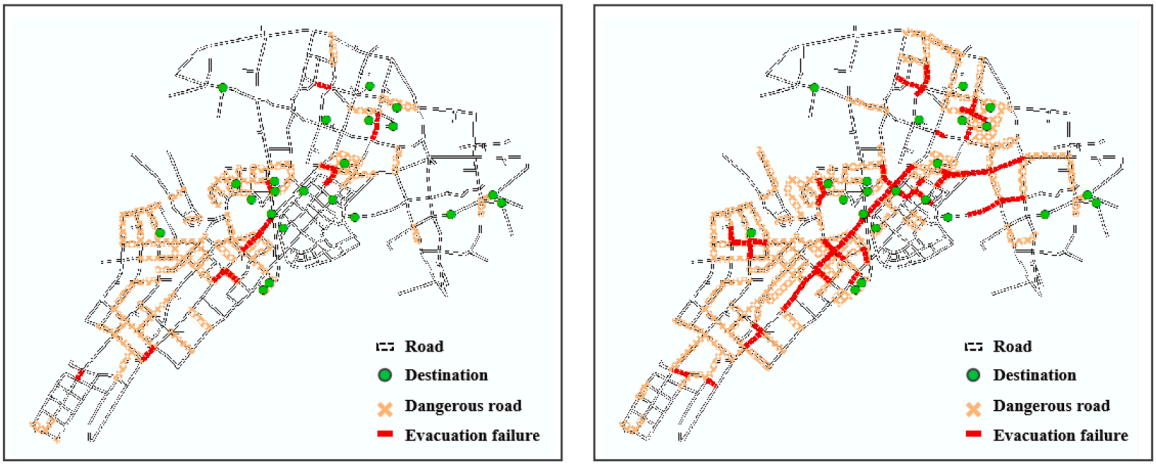
However, the reasons behind the low utilization rate of some shelters are different. For the shelter No. 17 and 18, the capacity of these two shelters in Table A2 is larger, which can accommodate more residents than the actual demand in reality. Therefore, the utilization rate is still low even if all residents have already evacuated successfully in the 10-year return period. For the shelter No. 9, it can be accessible in the early evacuation stage under the 10-year return period. However, the roads leading to the shelter become flooded with the increase of rainfall intensity, making it impassable for vehicles to reach under the 100-year return period. The above result underscores the importance of considering factors such as flooding conditions, shelter capacity, and geographical location when



(a) Without preparation time

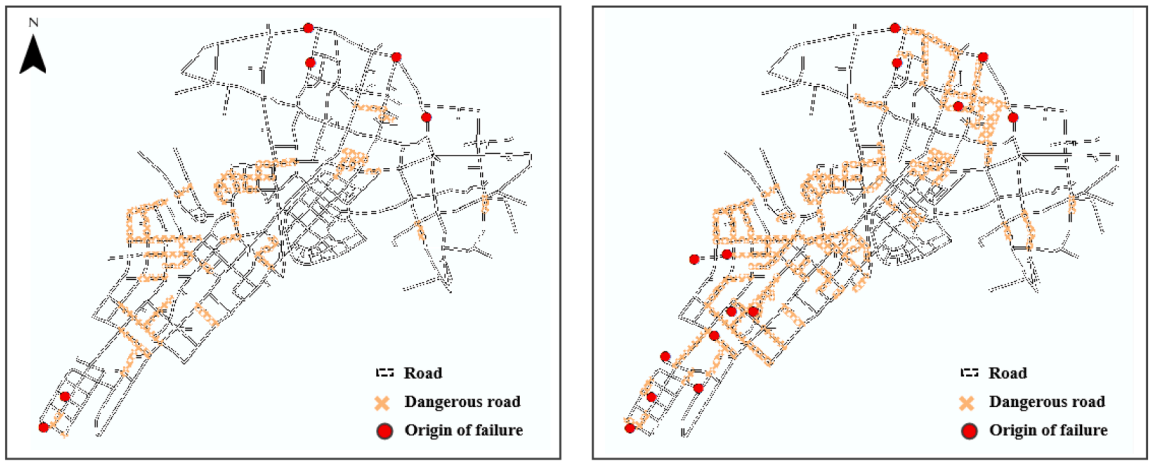


(b) 15 min preparation time

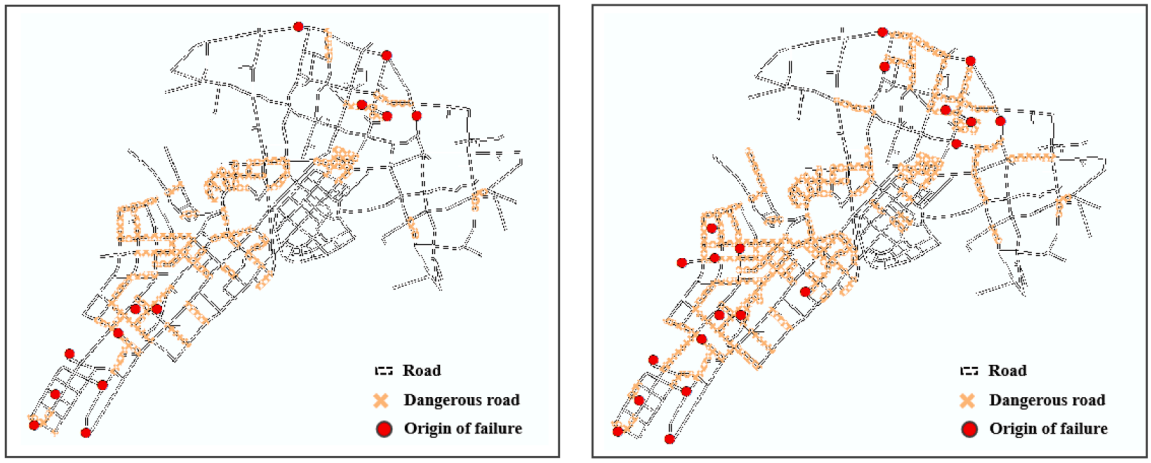


(c) 30 min preparation time

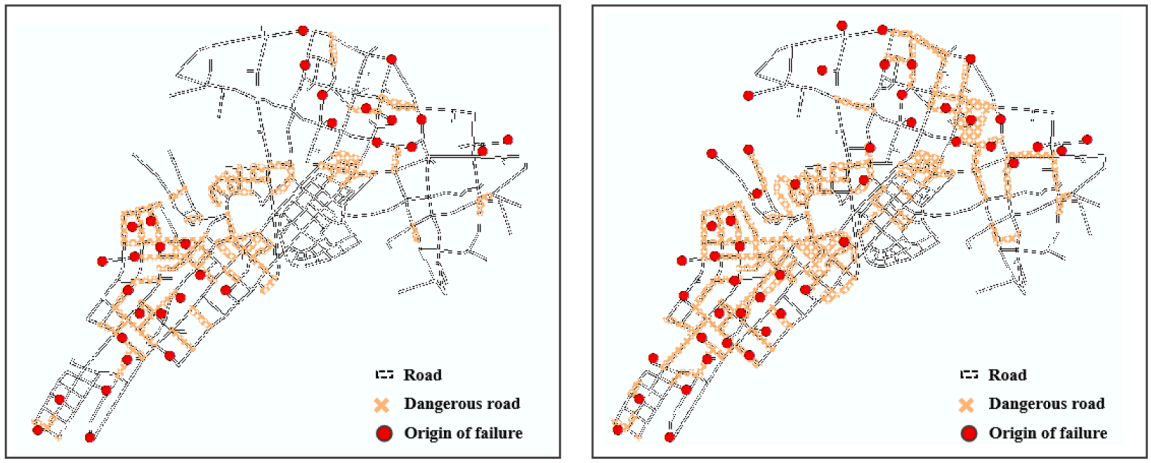
Fig. 13. Final location of evacuation failed vehicles. The left and right column represent the 10-year and 100-year return period respectively.



(a) Without preparation time

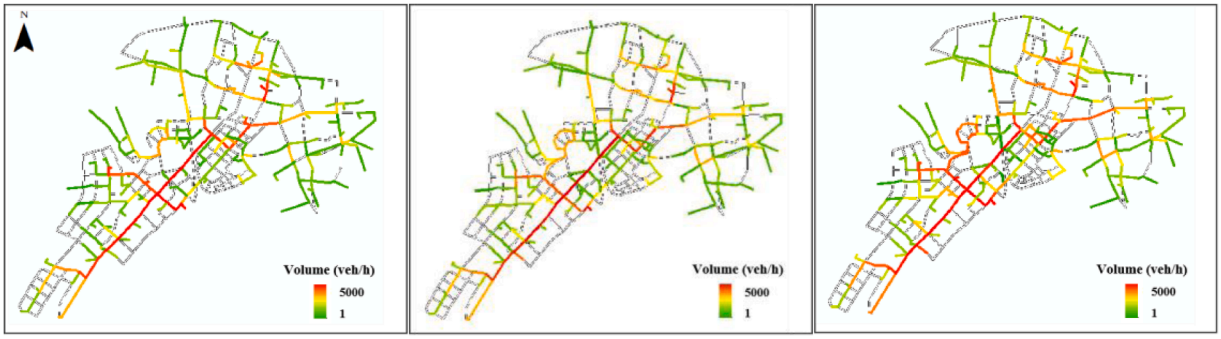


(b) 15 min preparation time

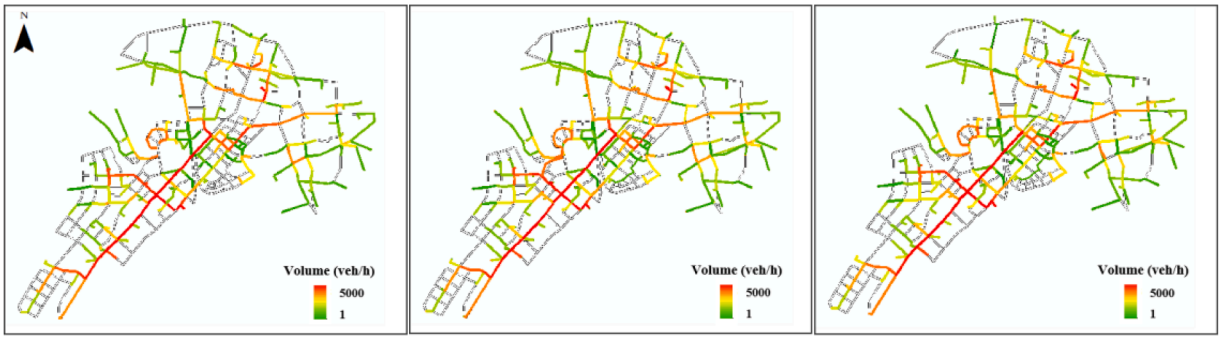


(c) 30 min preparation time

Fig. 14. Origins of evacuation failed vehicles. The left and right column represent the 10-year and 100-year return period respectively.



(a) 10-year return period



(b) 100-year return period

Fig. 15. Traffic volume distribution in different scenarios. The left, middle and right column represent the preparation time of 0 min, 15 min and 30 min respectively.

planning large scale evacuations in urban disaster events. These elements play a significant role in determining the success and efficiency of the evacuation process.

6. Conclusions and future work

Flood disasters are becoming increasingly frequent worldwide, leading to substantial property damage and casualties. In response to this growing threat, a dynamic evacuation planning framework is proposed for urban residents in flooding events. To achieve this goal, the study begins by simulating the flood inundation process using Mike models, providing critical data on flood depth and flow velocity across urban areas. Next, the impacts of floods on vehicle movement including travel speed and instability risk, and road conditions such as road capacity and accessibility are quantified. Then, the optimal evacuation routes based on evacuation time and evacuation risk are dynamically planned for residents in different areas. Finally, a case study is carried out in Nanjing Luhe District. The key findings of the study are summarized as follows.

(1) The proposed dynamic planning method considering both travel efficiency and movement safety outperforms other planning methods. It is found that the traditional shortest route plan without considering the flood impact is only efficient in the early evacuation stage. Although the performance of the shortest route plan with considering the flood risk is improved, the proposed plan can maximize the evacuation success rate, especially under extreme flood disasters.
(2) Both the rainfall intensity and preparation time have significant influences on evacuation outcomes. For the 30 min preparation time scenario, the success rate in the 100-year return period decreases by 10.9 % when compared to the 10-year return period scenario. It declines by 18.8 % and 24.6 % when compared to the immediate evacuation in the 10-year and 100-year return periods respectively. The evacuation failure analysis indicates that vehicles are difficult to move in the later evacuation stage due to the flood spreading, emphasizing the importance of minimizing the preparation time and carrying out efficient evacuation planning.
(3) The distribution of traffic volume on urban roads is uneven during the flood evacuation. The expressway with a large capacity carries the maximum traffic volume, which is vital for ensuring the evacuation efficiency. Meanwhile, many secondary roads connecting to shelters are also important to guarantee the ultimate success of evacuation. Maintaining clear access on these key roads during floods is essential.

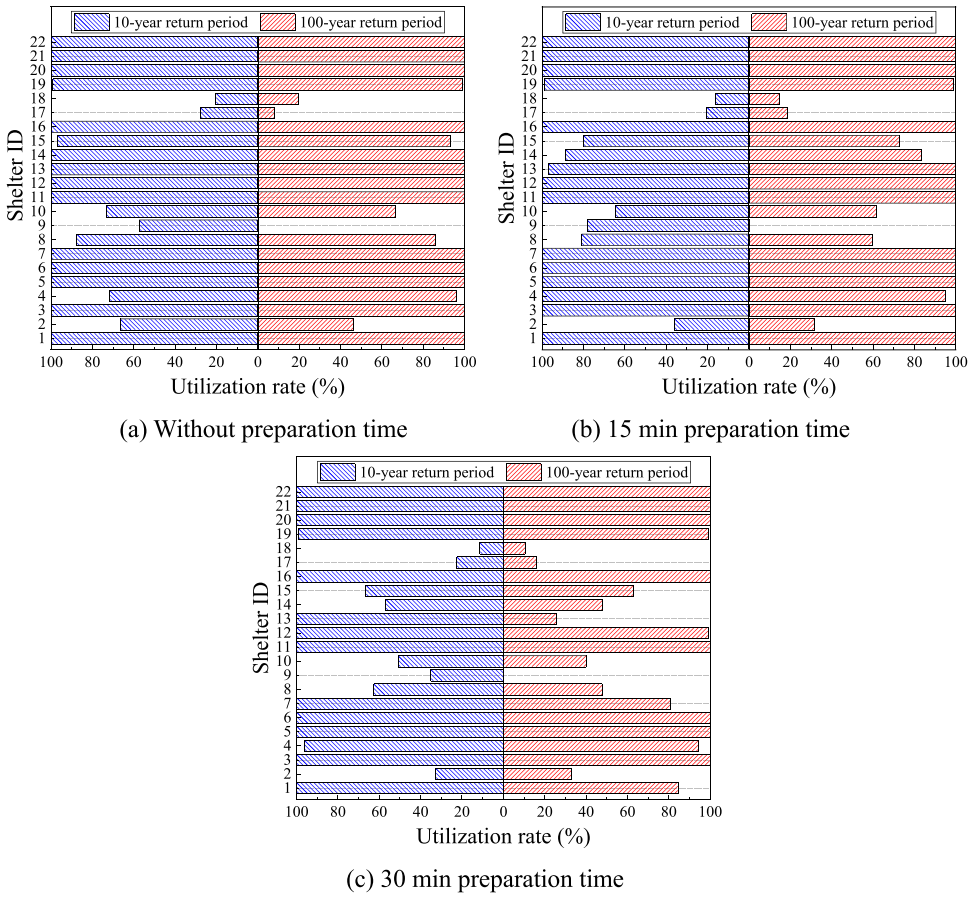


Fig. 16. Utilization rate of shelters under different preparation times.

(4) Significant variations in the utilization rate of shelters are found. Some shelters are fully utilized while others are underutilized or unoccupied. Besides the flooding blocking access to periphery roads of shelters, the geographical location and capacity of shelters can also affect their usage. These factors should be comprehensively considered in future urban planning.

The proposed methodology is helpful for relevant departments and urban residents in rapid decision-making, which can be used as a basis for making evacuation plans and evaluating management measures in urban flood disasters. The limitation of this work is as follows: First of all, only vehicle evacuation mode is considered, while residents will choose to evacuate through various transportation modes such walking, buses or automobiles in reality. Moreover, although the flood model has been validated through measured data, the evacuation results are not verified due to the lack of real large-scale evacuation data. In the future research, the multimodal evacuation should be explored based on population characteristics within the flooding area. In addition, it is necessary to collect real evacuation data under flood disasters to verify the reliability and robustness of the proposed route optimization model.

CRediT authorship contribution statement

Shuchao Cao: Conceptualization, Methodology, Funding acquisition, Supervision, Writing – original draft, Writing – review & editing. **Ying Zhou:** Formal analysis, Validation, Visualization, Writing – original draft. **Yunhe Tong:** Methodology, Writing – original draft. **Rui Ye:** Funding acquisition, Writing – review & editing. **Xiaoxia Yang:** Funding acquisition, Writing – review & editing. **Yiping Zeng:** Methodology, Writing – review & editing. **Peng Wang:** Methodology, Writing – review & editing.

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Appendix

Table A1
Number of residents in each residential point.

Residential ID	Population	Residential ID	Population	Residential ID	Population	Residential ID	Population
1	2026	26	654	51	505	76	1367
2	3353	27	1455	52	1944	77	1692
3	1853	28	617	53	3300	78	651
4	1913	29	77	54	1469	79	141
5	655	30	1630	55	513	80	435
6	2063	31	1442	56	670	81	2175
7	2104	32	1634	57	431	82	1177
8	1288	33	1704	58	1674	83	4021
9	1211	34	1782	59	2424	84	1893
10	471	35	939	60	1414	85	1671
11	1833	36	2476	61	1323	86	1882
12	3221	37	1238	62	2600	87	1890
13	1275	38	2220	63	937	88	2534
14	1307	39	967	64	2444	89	2253
15	768	40	970	65	1403	90	438
16	5915	41	1885	66	1434	91	606
17	1628	42	3105	67	1206	92	967
18	1245	43	1881	68	1695	93	277
19	681	44	1369	69	628	94	643
20	1092	45	1705	70	1232	95	4420
21	1684	46	1573	71	938	96	1023
22	653	47	1614	72	1640	97	4481
23	2006	48	1360	73	1504	98	1106
24	1071	49	1432	74	4435	99	1506
25	835	50	1696	75	3831	100	1805

Table A2
Population capacity in each shelter.

Shelter ID	Shelter name	Capacity (per)
1	Luhe Jinling Middle School Longhu Branch	10,000
2	Luhe District Jiangwan Primary School	5500
3	Luhe District Xiongzhou Junior High School	3840
4	Luhe District Wenshi Park	11,000
5	Luhe Lingyan Primary School	2600
6	Luhe Senior High School	4600
7	Luhe Experimental Senior High School	12,000
8	Luhe Lizhi School	13,000
9	Luhe Bilingual Primary School	6600
10	Luhe Citizen Center	33,000
11	Luhe Chengxi Junior High School	6000
12	Luhe Longchi Central Park	4000
13	Luhe Longchi Experimental Kindergarten	4000
14	Luhe Longchi Middle School	10,000
15	Luhe Longchi Primary School	20,000
16	Luhe District Mingde Primary School	2666
17	Luhe Longchi Lake East Leisure Plaza	14,000
18	Luhe Fenghuangshan Park	91,111
19	Luhe 805 Plaza	333
20	Luhe Airport East Road Roundabout Plaza	2000
21	Luhe Wanhui Plaza	3333
22	Luhe Longchi Lake Leisure Plaza	3333

Data availability

Data will be made available on request.

References

[1] G.M. Membele, M. Naidu, O. Mutanga, Examining flood vulnerability mapping approaches in developing countries: a scoping review, Int. J. Disaster. Risk. Reduct. 69 (2022) 102766.
[2] Y. Zhu, Y. Liu, L. Zhu, M. Liu, Z. Xie, J. Chen, X. Fu, Q. Wang, Z. Wang, Q. Du, Natural or man-made disaster? Lessons from the extreme rain and flood disaster in Zhengzhou, China on "2021.7.20", Sustainable Cities and Society, (2024) 105999.
[3] R. D'Ambrosio, A. Longobardi, Adapting drainage networks to the urban development: an assessment of different integrated approach alternatives for a sustainable flood risk mitigation in Northern Italy, Sustain. Cities. Soc. 98 (2023) 104856.
[4] G. Nearing, D. Cohen, V. Dube, M. Gauch, O. Gilon, S. Harrigan, A. Hassidim, D. Klotz, F. Kratzert, A. Metzger, S. Nevo, F. Pappenberger, C. Prudhomme, G. Shalev, S. Shenzis, T.Y. Tekalign, D. Weitzner, Y. Matias, Global prediction of extreme floods in ungauged watersheds, Nature 627 (2024) 559–563.
[5] Q. Li, J. Xia, M. Zhou, S. Deng, B. Dong, Risk assessment of metro tunnel evacuation in devastating urban flooding events, Tunn. Undergr .Space Technol. 144 (2024) 105540.
[6] Y. Zheng, X. Li, B. Jia, R. Jiang, Simulation of pedestrians' evacuation dynamics with underground flood spreading based on cellular automaton, Simul. Model. Pr. Theory 94 (2019) 149–161.
[7] G. Bernardini, E. Quagliarini, M. D'Orazio, M. Brocchini, Towards the simulation of flood evacuation in urban scenarios: experiments to estimate human motion speed in floodwaters, Saf. Sci. 123 (2020) 104563.
[8] S. Cao, M. Wang, G. Zeng, X. Li, Simulation of crowd evacuation in subway stations under flood disasters, IEEe trans. Intell. Transp. Syst. 25 (2024) 11858–11867.
[9] D. Shi, J. Li, Q. Wang, J. Chen, R. Lovreglio, J.T.Y. Lo, J. Ma, Modeling and simulation of crowd dynamics at evacuation bottlenecks during flood disasters, Transp. Res. E: Logist. Transp. Rev. 198 (2025) 104064.
[10] S. Cao, L. Ma, Y. Zeng, Y. Wang, P. Wang, L. Xu, Crowd safety assessment for flood evacuation in subway platform tunnels, Tunn. Undergr. Space Technol. 164 (2025) 106848.
[11] D. Shi, Q. Wang, J. Chen, R. Lovreglio, W. Yang, J. Ma, Exploring the influence of flooding conditions on crowd evacuation: an experimental study, Transp. Res. D: Transp. Environ. 146 (2025) 104868.
[12] X. Yang, W. Dai, Y. Li, X. Yang, An efficient evacuation path optimization for passengers in subway stations under floods, Tunn. Undergr. Space Technol. 143 (2024) 105473.
[13] W. Wang, Practical speed-flow relationship model of highway traffic-flow, J. Southeast Univ. Nat. Sci. Ed. 33 (2003) 487–491.
[14] M. Pregnolato, A. Ford, S.M. Wilkinson, R.J. Dawson, The impact of flooding on road transport: a depth-disruption function, Transp. Res. D: Transp. Environ. 55 (2017) 67–81.
[15] Y. Yang, M. Li, M. Xiong, J. Cao, Return period for urban rainwater drainage networks based on the lowest total social investment method: a case study in Tianjin, China, Pol. J. Environ. Stud. 28 (2019) 3993–3999.
[16] M.G.R. Fahad, R. Nazari, P. Bhavsar, M. Jalayer, M. Karimi, A decision-support framework for emergency evacuation planning during extreme storm events, Transp. Res. D: Transp. Environ. 77 (2019) 589–605.
[17] H. Alaeddine, K. Serrhini, M. Maizia, E. Néron, A spatiotemporal optimization model for the evacuation of the population exposed to flood hazard, Nat. Hazards Earth Syst. Sci. 15 (2015) 687–701.
[18] F.T. Zahura, J.L. Goodall, T.D. Chen, Using crowdsourced data to estimate passenger vehicle travel delays from nuisance flooding, Transp. Res. D: Transp. Environ. 133 (2024) 104307.
[19] K. He, M. Pregnolato, N. Carhart, J. Neal, R. De Risi, Functionality assessment of road network combining flood roadworthiness and graph topology, Transp. Res. D: Transp. Environ. 135 (2024) 104354.
[20] C. Li, C. Jiang, J. Chen, M.Y. Lam, J. Xia, R. Ahmadian, An overview of flood evacuation planning: models, methods, and future directions, J. Hydrol. 656 (2025) 133026.
[21] C. Arrighi, H. Oumeraci, F. Castelli, Hydrodynamics of pedestrians' instability in floodwaters, Hydrol. Earth. Syst. Sci. 21 (2017) 515–531.
[22] G. Musolino, R. Ahmadian, J. Xia, R.A. Falconer, Mapping the danger to life in flash flood events adopting a mechanics based methodology and planning evacuation routes, J. Flood Risk Manag. 13 (2020) 12627.
[23] Z. Zhu, Y. Zhang, L. Gou, D. Peng, B. Pang, On the physical vulnerability of pedestrians in urban flooding: experimental study of the hydrodynamic instability of a human body model in floodwater, Urban. Clim. 48 (2023) 101420.
[24] B. Evans, A. Lam, C. West, R. Ahmadian, S. Djordjević, A. Chen, M. Pregnolato, A combined stability function to quantify flood risks to pedestrians and vehicle occupants, Sci. Total. Environ. 908 (2024) 168237.
[25] F.Y. Teo, J. Xia, R.A. Falconer, B. Lin, Experimental studies on the interaction between vehicles and floodplain flows, Int. J. River Basin Manag. 10 (2012) 149–160.
[26] J. Xia, R.A. Falconer, X. Xiao, Y. Wang, Criterion of vehicle stability in floodwaters based on theoretical and experimental studies, Nat. Hazards 70 (2014) 1619–1630.
[27] C. Arrighi, M. Pregnolato, R.J. Dawson, F. Castelli, Preparedness against mobility disruption by floods, Sci. Total. Environ. 654 (2019) 1010–1022.
[28] S.M.H. Shah, Z. Mustaffa, E. Martinez-Gomariz, D.K. Kim, K.W. Yusof, Criterion of vehicle instability in floodwaters: past, present and future, Int. J. River Basin Manag. 19 (2021) 1–23.
[29] W. Lv, F. Deng, J. Wang, Y. Han, S. Yang, Large-scale real-time evacuation modeling during urban floods: a coupled agent-based multi-model framework, Simul. Model. Pr. Theory 140 (2025) 103075.
[30] M. He, C. Chen, F. Zheng, Q. Chen, J. Zhang, H. Yan, Y. Lin, An efficient dynamic route optimization for urban flooding evacuation based on cellular automata, Comput. Environ. Urban Syst. 87 (2021) 101622.
[31] M. Borowska-Stefańska, L. Balážovičová, K. Goniewicz, M. Kowalski, P. Kurzyk, M. Masný, S. Wiśniewski, M. Žoncová, A. Khorram-Manesh, Emergency management of self-evacuation from flood hazard areas in Poland, Transp. Res. D: Transp. Environ. 107 (2022) 103307.
[32] T. Mizumura, H. Taguchi, H. Nakamura, Confirming the safety improvement and evacuation time reduction effects by the method for evacuating inundated areas via the shortest possible route, Int. J. Disaster. Risk. Reduct. 101 (2024) 104252.
[33] M. Yazdani, M. Mojtahedi, M. Loosemore, D. Sanderson, V. Dixit, An integrated decision model for managing hospital evacuation in response to an extreme flood event: a case study of the Hawkesbury-Nepean River, NSW, Australia, Saf. Sci. 155 (2022) 105867.
[34] M. Yazdani, M. Haghani, Elderly people evacuation planning in response to extreme flood events using optimisation-based decision-making systems: a case study in western Sydney, Australia, Knowl. Based. Syst. 274 (2023) 110629.
[35] Z. Han, Y. Mitani, K. Kawano, H. Taniguchi, H. Honda, L. Meng, Z. Li, Quantitative assessment of flooding risk based on predicted evacuation time: a case study in Joso city, Japan, Int. J. Disaster. Risk. Reduct. 98 (2023) 104113.
[36] W. Yang, X. Yan, D. Hu, X. Jiang, J. Song, R. Lang, K. Xu, J. Lian, A novel emergency evacuation route optimization model in flood disasters using hydrodynamic model and intelligent algorithm, Saf. Sci. 182 (2025) 106709.
[37] W. Li, K. Lin, T. Zhao, T. Lan, X. Chen, H. Du, H. Chen, Risk assessment and sensitivity analysis of flash floods in ungauged basins using coupled hydrologic and hydrodynamic models, J. Hydrol. 572 (2019) 108–120.
[38] A. Balakrishna Madayala, A. Jain, B. Lohani, Development of a physics-guided neural network model for effective urban flood management, J. Hydrol. Eng. 27 (2022) 04022017.
[39] R. Zhou, H. Zheng, Y. Liu, G. Xie, W. Wan, Flood impacts on urban road connectivity in southern China, Sci. Rep. 12 (2022) 16866.
[40] O.F. Abioye, M.A. Dulebenets, E.E. Ozguven, R. Moses, W.R. Boot, T. Sando, Assessing perceived driving difficulties under emergency evacuation for vulnerable population groups, Socioecon. Plann. Sci. 72 (2020) 100878.

[41] L. Du, X. Yang, Research on the impact of different road waterlogging conditions on traffic and the resulting losses, in: Innovation in Science and Technology for Green Transportation-The 11th National Conference on Urban Road Traffic. Dalian, Liaoning, China, 2011, p. 5.
[42] E. Martínez-Gomariz, M. Gómez, B. Russo, S. Djordjević, A new experiments-based methodology to define the stability threshold for any vehicle exposed to flooding, Urban. Water. J. 14 (2017) 930–939.
[43] TRB, Highway Capacity Manual 2010, Transportation Research Board, Washington D.C., 2010.
[44] N. Wang, J. Hou, Y. Du, H. Jing, T. Wang, J. Xia, J. Gong, M. Huang, A dynamic, convenient and accurate method for assessing the flood risk of people and vehicle, Sci. Total. Environ. 797 (2021) 149036.
[45] J. Zhang, F. Jiang, D. Peng, Research on impact of traffic flow in rainy environments, China Saf. Sci. J. 33 (2023) 134–143.
[46] C.J. Keifer, H.H. Chu, Synthetic storm pattern for drainage design, J. Hydraul. Div. 83 (1957), 1332-1331-1332-1325.
[47] Statistical Yearbook of Nanjing City, in. (Nanjing Municipal Bureau of Statistics, 2024).
[48] MOT, Technical Standard of Highway Engineering(JTG B01-2014, China Communications Press, Beijing, China, 2014, pp. 5–10.
[49] R. Sreejith, K.R. Sinimole, Modelling evacuation preparation time prior to floods: a machine learning approach, Sustain. Cities. Soc. 87 (2022) 104257.